THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 1 & 2: **Orienting Yourself: The Use of Coordinates, Introduction to Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Quadrant III	signs of (x, y) are $(-, -)$
2	**(a)** 5 units	$\sqrt{(4-(1))^2+(6-(2))^2} = \sqrt{9+16} = 5$
3	**(a)** 10 units	$\sqrt{(4-(-4))^2+(3-(-3))^2} = \sqrt{64+36} = 10$
4	**(b)** $(2, 2)$	$\left(\frac{-3+(7)}{2}, \frac{5+(-1)}{2}\right)$
5	**(d)** $(5, 8)$	$B = (2x_M - x_A,\ 2y_M - y_A)$
6	**(a)** 0	highest power of the variable is 0
7	**(d)** -9	$2(-6)+(3) = -9$
8	**(a)** -11	$-2(5)+(-1) = -11$
9	**(c)** 4	substitute $x = -1$ to get 4
10	**(d)** $6n - 5$	common difference 6, so $a_n = 6n - 5$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$O(0,0),\ A(7,0),\ B(7,4),\ C(0,4)$	2
12	$(-2,5)$: Quadrant II; $(6,-3)$: Quadrant IV; $(0,-4)$: on the y-axis; $(-1,-1)$: Quadrant III	½ × 4
13	Amount $= 200 + 50m$. $200 + 50m = 750 \Rightarrow m = 11$ matches.	1+1
14	$y = 2x + 3$; it cuts the y-axis at $(0, 3)$.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$RB = \sqrt{9+16} = 5$; $BC = \sqrt{36+49} = \sqrt{85}$; $RC = \sqrt{81+121} = \sqrt{202}$. $(RB+BC)^2 = 25+85+10\sqrt{85} = 110+10\sqrt{85} \neq 202 = RC^2$, so $RB + BC \neq RC$. Hence the points are not collinear.	1+1+1
16	$b(x) = 600 - 15x$; zero when $x = 40$ days. Negative slope ⇒ linear decay.	2+1
17	$A'(-3,4),\ D'(-7,1),\ M'(-9,6)$. $AD = \sqrt{16+9} = 5$; $A'D' = \sqrt{(-7+3)^2+(1-4)^2} = \sqrt{16+9} = 5$. Reflection preserves lengths.	1+1+1
	OR: Let $P(0,y)$. $PA^2 = 36+(y-5)^2$, $PB^2 = 16+(y-3)^2$. $36+y^2-10y+25 = 16+y^2-6y+9 \Rightarrow 61-10y = 25-6y \Rightarrow y = 9$. $P = (0,9)$	1+1+1
18	Intercepts -1, 1, 5. All three have slope 2, so they are parallel (equal slopes, different intercepts).	1+1+1
	OR: Points: $(0,0),(4,2)$; $(0,0),(2,2)$; $(0,0),(2,4)$. All pass through the origin; larger a ⇒ steeper line; $a = 1$ makes equal angles with both axes.	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$AB^2 = 16+9 = 25$; $BC^2 = 49+1 = 50$; $AC^2 = 9+16 = 25$. $AB^2 + AC^2 = BC^2$ ⇒ right angle at A. Area $= \frac{1}{2} \times 5 \times 5 = 12.5$ sq. units.	3+2
	OR: (i) $AB = \sqrt{36+64} = 10$ ⇒ 1000 m. (ii) $BC = 8$; total $= 18$ units $= 1800$ m. (iii) $AC = 6$ ⇒ saves $18 - 6 = 12$ units $= 1200$ m.	2+2+1
20	(i) $x + 4x = 300 \Rightarrow$ 60 ft and 240 ft. (ii) $2(3w+3) = 24 \Rightarrow w = 3$ m, $l = 9$ m. (iii) 27 m^2.	2+2+1
	OR: Points: $(0,0),(1,3)$ and $(0,0),(1,-3)$. (i) At the origin. (ii) Slopes 3 and -3; same steepness, opposite directions. (iii) Positive slope ⇒ growth; negative slope ⇒ decay.	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 1 & 2: **Orienting Yourself: The Use of Coordinates, Introduction to Linear Polynomials** | Class 9 Maths |
Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 13 units	$\sqrt{25+144}=13$
2	**(b)** $(2,\ 6)$	reflection in the y-axis changes the sign of the x-coordinate
3	**(b)** 0	every point on the x-axis has $y=0$
4	**(c)** $(4,\ -4)$	Quadrant II needs $x<0,\ y>0$
5	**(b)** 4 units	distance from the x-axis is $\lvert y\rvert$
6	**(c)** 4 and -7	compare with $y=ax+b$
7	**(d)** lies on the line	RHS at $x=7$ is 15, equal to 15
8	**(a)** $5x-8$	degree 1
9	**(d)** parallel	equal slopes, different intercepts
10	**(b)** False, its degree is 2 so it is quadratic	degree of $5x^2-3$ is 2

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$AB=\sqrt{4+4}=2\sqrt{2},\ BC=2\sqrt{2},\ AC=\sqrt{16+16}=4\sqrt{2}$. Since $AB+BC=AC$, the points are collinear.	1+1
12	$\frac{-4+x}{2}=2\Rightarrow x=8;\ \frac{3+y}{2}=-1\Rightarrow y=-5$. So $B=(8,\ -5)$	1+1
13	Slope $=-4$, y-intercept $=9$; negative slope ⇒ linear decay.	1+1
14	$h(5)=3-2.5=0.5$ m. Empty when $3-0.5t=0\Rightarrow t=6$ months.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Distances: $\sqrt{16+16}=4\sqrt{2}$ from each. The three points form an isosceles triangle with the apex at $(0,-1)$; base $=8$ and $64=32+32$, so it is also right-angled at the apex.	1+1+1
16	For $x=0,1,2$: $(0,-2),\ (1,1),\ (2,4)$. It cuts the y-axis at $(0,-2)$.	2+1
17	Radius $=\sqrt{1+64}=\sqrt{65}$ (check $16+49=65,\ 49+16=65$ ✓). $OD^2=25+36=61<65$ ⇒ $(-5,6)$ lies **inside** the circle.	1+1+1
	OR: Diagonals bisect each other: midpoint of $AC=(3.5,4)$. If $D=(x,y)$ then $\frac{4+x}{2}=3.5\Rightarrow x=3,\ \frac{3+y}{2}=4\Rightarrow y=5$. $D=(3,5)$	1+1+1
18	$y=-2x$ (each y is -2 times x). Slope -2 ⇒ the line falls; y decreases by 2 for each unit increase in x (linear decay).	1+1+1
	OR: $A=13b$: 156, 130, 104 cm^2. The area changes by $13\times$ (change in b); a linear pattern with constant difference 26 for a decrease of 2 cm in b.	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sides: $\sqrt{9+81}=\sqrt{90}$ each (check $\sqrt{81+9}=\sqrt{90},\ \sqrt{9+81},\ \sqrt{81+9}$). Diagonals: $\sqrt{144+144}=12\sqrt{2}$ and $\sqrt{36+36}=6\sqrt{2}$. Area $=\frac{1}{2}\times 12\sqrt{2}\times 6\sqrt{2}=72$ sq. units.	3+2
	OR: Let $P(x,0)$: $(x+2)^2+25=(x-2)^2+9\Rightarrow 8x=-16\Rightarrow x=-2$. $P(-2,0)$; distance $=\sqrt{0+25}=5$.	3+2
20	(i) $b=5$; $13=4a+5\Rightarrow a=2$. (ii) $y=2x+5$. (iii) $2(10)+5=25$ ✓ yes. (iv) $y=0\Rightarrow x=-2.5$: $(-2.5,0)$.	2+1+1+1
	OR: (i) $20+5x=70\Rightarrow x=10$. (ii) $C(n)=20+10n$. (iii) ₹140. (iv) $20+10n=170\Rightarrow n=15$.	1+1+1+2

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ANSWER KEY & MARKING SCHEME SET C

Chapters 1 & 2: **Orienting Yourself: The Use of Coordinates, Introduction to Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** Quadrant I	signs of (x, y) are $(+, +)$
2	**(c)** 13 units	$\sqrt{(3-(-2))^2+(-9-(3))^2} = \sqrt{25+144} = 13$
3	**(a)** 17 units	$\sqrt{(10-(2))^2+(14-(-1))^2} = \sqrt{64+225} = 17$
4	**(d)** $(3,\ -4)$	$\left(\frac{0+(6)}{2}, \frac{-8+(0)}{2}\right)$
5	**(d)** $(-7,\ 2)$	$B = (2x_M - x_A,\ 2y_M - y_A)$
6	**(a)** 1	highest power of the variable is 1
7	**(d)** 13	$3(4) + (1) = 13$
8	**(d)** 6	substitute $x = 0$ to get 6
9	**(c)** $4n - 1$	common difference 4, so $a_n = 4n - 1$
10	**(a)** $5n + 2$	common difference 5, so $a_n = 5n + 2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Distance of $(-5, -12)$ is $\sqrt{25+144} = 13$; distance of $(13, 0)$ is 13. Yes, both lie on the circle of radius 13 centred at the origin.	1+1
12	$AB = \sqrt{4+4} = 2\sqrt{2},\ BC = 2\sqrt{2},\ AC = 4$. $AB = BC$, and $AB^2 + BC^2 = 8 + 8 = 16 = AC^2$: a right-angled isosceles triangle.	1+1
13	$2(7) + 1 = 15$ ✓. $2(3) + 1 = 7 \neq 8$ ×.	1+1
14	$2n - 1$. $2n - 1 = 47 \Rightarrow n = 24$.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$GL = \sqrt{64+36} = 10$ units; midpoint $= (4, 3)$.	2+1
16	$V = 77h$: 385, 693, 1001 cm^3. Consecutive differences are equal (308) — a linear pattern.	2+1
17	Sides: 4, 3 and $\sqrt{16+9} = 5$. Perimeter $= 12$ units.	2+1
	OR: Fourth vertex $(2, 5)$. Diagonal $= \sqrt{36+16} = 2\sqrt{13}$ units.	1+2
18	$500 - 20d$; zero when $d = 25$ days. After 15 days, 200 pages remain.	2+1
	OR: First month ₹500; second ₹650… $a_n = 500 + 150(n-1) = 350 + 150n$. $350 + 150n = 1850 \Rightarrow n = 10$.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$AB = 4$, $BC = \sqrt{4+4} = 2\sqrt{2}$, $CA = 2\sqrt{2}$; $PQ = 8$, $QR = \sqrt{16+16} = 4\sqrt{2}$, $RP = 4\sqrt{2}$. Ratios $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP} = \frac{1}{2}$ $\Rightarrow$ similar.	3+2
	OR: $(k-3)^2 + 36 = 100 \Rightarrow (k-3)^2 = 64 \Rightarrow k = 11$ or $k = -5$. Points $(11, 2)$ and $(-5, 2)$.	3+2
20	(i) $x = 1$: 9; $x = -2$: $20 + 6 + 7 = 33$. (ii) e.g. $x^3 - 7x^2 + 2x + 1$. (iii) Degree 3; coefficient of z is 0 (no z term).	2+2+1
	OR: (i) $A = 500 + 150n$. (ii) ₹1,400. (iii) $500 + 150n = 2000 \Rightarrow n = 10$. (iv) A straight line through $(0, 500)$ and $(10, 2000)$; slope 150 = monthly addition.	1+1+1+2

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ANSWER KEY & MARKING SCHEME SET D

Chapters 1 & 2: **Orienting Yourself: The Use of Coordinates, Introduction to Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 15 units	$\sqrt{81+144}=15$
2	**(a)** $(-5,\ 7)$	reflection in the y-axis changes the sign of the x-coordinate
3	**(c)** $(0,\ -5)$	$x=0$, and below the origin means $y<0$
4	**(c)** are collinear	O is the midpoint of AB
5	**(a)** $(3,\ 0)$	a point on the axis is its own reflection
6	**(d)** 1 and 3	compare with $y=ax+b$
7	**(a)** does not lie on the line	RHS at $x=2$ is 4, not equal to 5
8	**(c)** 6	read off the term $6x^2$
9	**(c)** positive slope	values increase with x
10	**(d)** 10 cm	$4x=10$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(0,6)$ and $(-4,0)$; distance $=\sqrt{16+36}=\sqrt{52}=2\sqrt{13}$ units.	1+1
12	$PA^2=1+9=10$; $PB^2=9+1=10$. Yes, P is equidistant.	1+1
13	$25+15\times 8=$ ₹145.	1+1
14	$x+(x+10)=64\Rightarrow x=27$; numbers 27 and 37.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$P=\left(\frac{2(2)+8}{3},\frac{2(-2)+4}{3}\right)=(4,0)$. $AP=\sqrt{4+4}=2\sqrt{2}$; $AB=\sqrt{36+36}=6\sqrt{2}$; $AP=\frac{1}{3}AB$ ✓	2+1
16	Width $=10-x$; area $=x(10-x)=10x-x^2$ (a quadratic polynomial). At $x=6$: $60-36=24$ cm^2.	2+1
17	Right angle at O. Hypotenuse $AB=\sqrt{36+64}=10$ units (1½). Area $=\frac{1}{2}\times 6\times 8=24$ sq. units (1½).	1½+1½
	OR: $AB=\sqrt{9+16}=5$, $AC=\sqrt{16+9}=5$, $BC=\sqrt{49+1}=\sqrt{50}$. $AB=AC$ (isosceles) and $AB^2+AC^2=50=BC^2$ (right angle at A).	1+1+1
18	$y=3,5,11,15,19,25,41$. $(4,9)$: $2(4)+1=9$ ✓ lies on the line; $(6,14)$: $2(6)+1=13\neq 14$ ×.	2+1
	OR: $3x=15\Rightarrow x=5$; check $3(5)-4=11$ ✓. Example: the sum of two numbers is 64 and one is 10 more than the other: $x+(x+10)=64\Rightarrow 2x+10=64$.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $F=(6,4)$. (ii) $FL_1=6=FL_2$ ✓. (iii) Area $=96$ sq. units; $OB=\sqrt{144+64}=\sqrt{208}=4\sqrt{13}$ units.	1+2+2
	OR: $M=(2.5,4)$. $CM=\sqrt{(7-2.5)^2+(2-4)^2}=\sqrt{20.25+4}=\frac{\sqrt{97}}{2}\approx 4.92$ units.	2+3
20	$3=a+b$, $7=3a+b\Rightarrow a=2$, $b=1$: $y=2x+1$. (i) $2(5)+1=11$ ✓. (ii) 1. (iii) $y=2x-1$.	2+1+1+1
	OR: (i) $200l+160w+50lw$. (ii) $4000+1600+10000=$ ₹$15,600$. (iii) Terms: $200l$, $160w$, $50lw$; variables l, w; coefficients 200, 160, 50.	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.